

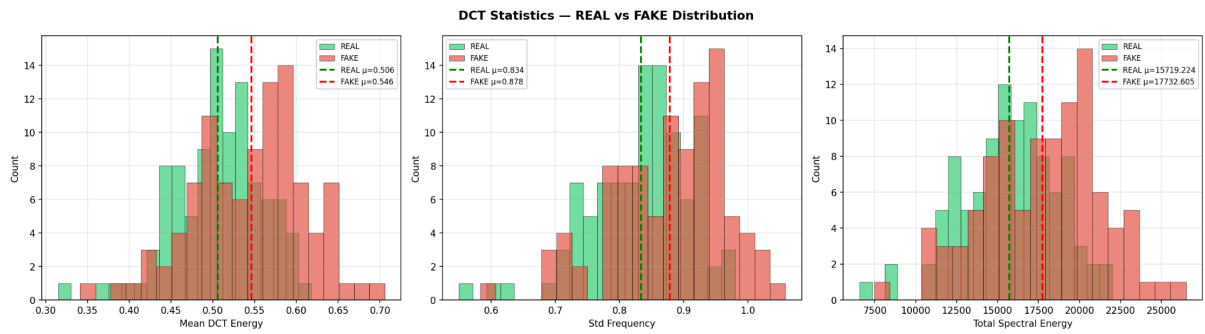
Phrase 4 and 5

Table 1. DCT Statistics Comparison

Metric	Real Images	Fake Images	Change (Fake vs. Real)
Mean Energy	0.5055	0.5463	+8.1%
Standard Frequency	0.8336	0.8784	+5.4%
Total Energy	15,719.2236	17,732.6055	+12.8%

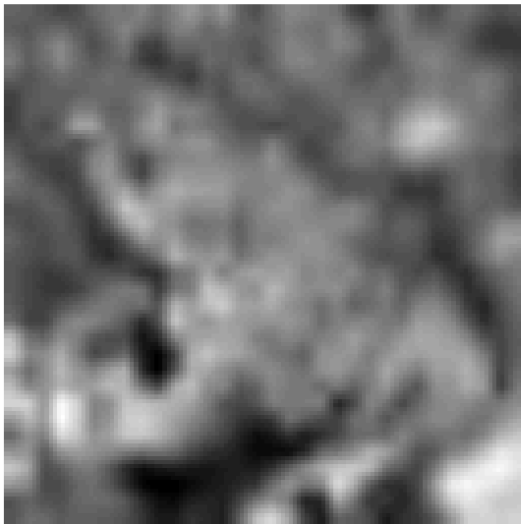
Table 2. DCT Analysis Interpretation

Observation	Interpretation
Mean Energy	Fake images exhibit higher mean spectral energy than real images.
Frequency Variation	Fake images show greater frequency variation, indicating more irregular high-frequency components.
Total Energy	The overall DCT energy is significantly higher in fake images (+12.8%).
Key Finding	AI-generated images contain measurable frequency-domain artifacts that are not easily observable in the spatial domain.
Conclusion	These quantitative differences provide strong evidence for incorporating frequency-domain (DCT) features into the deepfake detection framework.

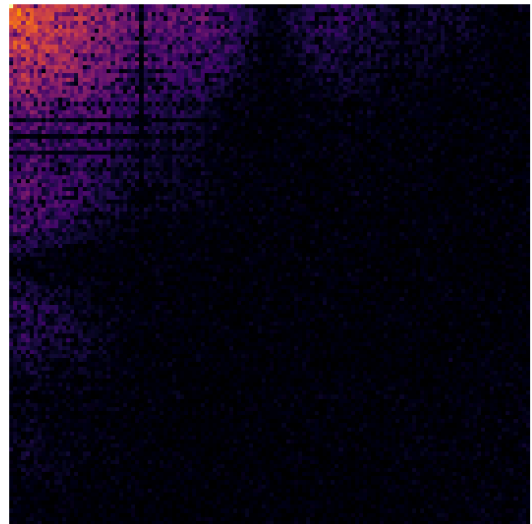


Spatial vs DCT Frequency Spectrum — REAL (top) / FAKE (bottom)

REAL — Spatial (Grayscale)



REAL — DCT Frequency Spectrum



FAKE — Spatial (Grayscale)



FAKE — DCT Frequency Spectrum

